your brain is not any more independent from what is happening in the atoms of mine: we communicate.

The world isn't, then, just a network of colliding atoms: it is also a network of correlations between sets of atoms, a network of real reciprocal information between physical systems.

In all of this, there is nothing idealistic or spiritual; it's nothing but an application of Shannon's idea that alternatives can be counted. All this is as much a part of the world as the stones of the Dolomites, the buzzing of bees and the waves of the sea.

Once we have understood that this network of reciprocal information exists in the universe, it is natural to seek to use this treasure to describe the world. Let's start with an aspect of nature well understood since the end of the nineteenth century: heat. What *is* heat? What does it mean to say that something is hot? Why does a cup of scalding-hot tea cool itself down, rather than heating itself up further?

It was the Austrian scientist Ludwig Boltzmann, the founder of statistical mechanics, who first understood why.[fn49] Heat is the random microscopic movement of molecules: when the tea is hotter, the movement of the molecules is more agitated. Why does it cool down? Boltzmann hazarded a splendid hypothesis: because the number of possible states of the molecules in hot tea and cold air is smaller than the number in cool tea and slightly warmer air. The combined state evolves from a situation where there are less possible states to a situation where there are more possible states. The tea can't warm itself up, because information cannot increase by itself.

I'll elaborate. The molecules of tea are extremely numerous and extremely small, and we don't know their precise movements. Therefore, we lack information. This lack of information – or missing information – can be computed. (Boltzmann did it: he computed the number of distinct states the molecules can be in. This number depends on the temperature.) If the tea cools, a little of its energy passes into the surrounding air; therefore, the molecules of tea move more slowly and the molecules of air move more quickly. If you compute your missing information, you discover that it has increased. If, instead, tea absorbed heat from the colder air, then the missing information would be decreased. That is, we would know more. But information cannot fall from the sky. It cannot increase by itself, because what we don't know, we just don't know. Therefore, the tea cannot warm up by itself in contact with cold air. It sounds a bit magical, but it works: we can predict how heat behaves just on the basis of the observation that our information cannot increase for free!

Boltzmann was not taken seriously. At the age of fifty-six, in Duino, near Trieste, he committed suicide. Today, he is considered one of the geniuses of physics. His tomb is incised with his formula